

MAINTENANCE OPERATIONS CLIENT

END-TO-END Project Report

Business Analysis | Data Analysis | Power BI Reporting

Project Summary	Detail
Period	Aug-Nov 2024
Role	Business Analyst + Data Analyst
Primary Stakeholder	Maintenance Manager
Data Scope	Approximately 10,000 historical records across 20 teams
Solution	Power BI dashboard built from scratch
Confirmed KPI	92% overall completion rate
Confirmed Impact	Approximately 3-4 hours of weekly manual reporting effort eliminated

1. Business Problem

Maintenance reporting relied on manual consolidation of operational updates, making it difficult to track completion, overdue work and team performance efficiently. The project focused on structuring fragmented maintenance data and creating a reusable reporting view for the Maintenance Manager.

2. Data Landscape

Area	Confirmed Scope
Sources	Excel, CSV, Google Sheets and database
Records	Approximately 10,000 historical records
Operational scope	Approximately 20 teams
Records covered	Maintenance tasks, work orders, inspections, service requests and equipment records
Key fields	Work Order/Task ID, Team, Technician, Asset/Equipment, Created Date, Due Date, Completion Date, Status, Priority, Category, Location, Remarks
Quality issues	Duplicates, missing values, inconsistent team names, inconsistent date formats, inconsistent status names and blank records

3. Project Approach

Stage	Work Performed
Requirements	Translated maintenance reporting needs into KPI, exception and operational-analysis requirements.
Data preparation	Cleaned and structured the historical records and addressed known quality issues.
Business rules	Defined governed logic for key measures, including completion and overdue work.
Analysis	Analyzed completion, work status, overdue work, completion time, priority, teams, technicians, backlog, workload, categories and equipment.
Reporting	Built the Power BI dashboard from scratch for reusable operational visibility.
Validation	Validated data quality, KPI logic, filters and dashboard reconciliation.
Delivery	Provided dashboard, documentation, presentation, live walkthrough and training.

4. Core Business Rules & KPIs

Measure	Definition / Scope
Completion Rate	Completed Maintenance Tasks / Total Assigned Maintenance Tasks x 100
Observed Completion Rate	92%
Overdue	Due Date < Current Date AND Status != Completed
Work Status	Completed, open and pending work were analyzed
Average Completion Time	Analyzed; exact historical value/calculation convention not retained
Backlog	Confirmed analytical area; exact historical definition not retained
Priority	Priority breakdown and high-priority pending work
Team / Technician	Completion and performance analysis within operational context
Category / Asset	Recurring maintenance categories and equipment-related patterns

KPI boundary: The 92% completion rate was surfaced by the analysis. It is not a 92% improvement and is not claimed as an increase caused by the project.

5. Key Findings

Finding	Interpretation
92% completion rate	Established the overall maintenance completion position.
Backlog	Outstanding maintenance workload was identified.
Overdue work	Late incomplete work was visible under the confirmed overdue rule.
Team differences	Performance varied across teams.
Workload imbalance	Uneven workload distribution was identified.
High-priority pending work	Critical outstanding work required management visibility.
Recurring categories	Repeated maintenance categories were identified.
Equipment patterns/issues	Equipment-related patterns were identified for further operational attention.

6. Power BI Solution

The Power BI solution consolidated maintenance performance and exception reporting into a management view covering completion, work status, overdue items, priority, teams, technicians, backlog, categories and assets/equipment.

Portfolio Dashboard Area	Purpose
Executive Maintenance Overview	Overall operational position and exception visibility
Work Order & Completion Analysis	Work status, completion and overdue analysis
Team & Technician Performance	Performance and workload comparison
Backlog, Priority & Asset Analysis	Outstanding work, criticality and equipment/category patterns

Dashboard structure: These page names are the portfolio structure; the exact historical dashboard page names are not retained.

7. Findings to Action

Recommendations were made and implemented during the engagement, but the exact historical recommendation wording is not retained. For portfolio purposes, the confirmed findings support management focus on overdue work, high-priority pending items, workload distribution, recurring maintenance activity and equipment-level investigation.

8. Testing & Validation

Validation Area	Control
Data quality	Duplicates, missing values, naming inconsistencies, dates and blank records
Completion	Reconciliation to the governed completion formula
Overdue	Every overdue record must satisfy the confirmed rule
Filters	Relevant visuals respond consistently to operational filters
Cross-page reconciliation	Equivalent measures reconcile under equivalent filter context
Confidentiality	Public portfolio content does not expose confidential client/personnel information

9. Business Outcome

Reporting efficiency: Approximately 3-4 hours of weekly manual reporting effort were eliminated.

The project improved reporting efficiency by replacing recurring manual consolidation with a reusable Power BI reporting solution. It also gave the Maintenance Manager clearer visibility into overall completion and operational exceptions.

10. Tools Used

Area	Tools
Data preparation / analysis	Excel, SQL, Python, Power Query, Google Sheets
Reporting	Power BI
Measure layer	DAX
Business analysis	Requirements definition, KPI/business-rule definition, operational analysis and validation

11. Portfolio Evidence Boundaries

Confirmed	Not Claimed
Approximately 10,000 historical records across 20 teams	Exact confidential client identity or location
92% overall completion rate	Completion improvement caused by the project
Confirmed overdue rule	Exact backlog formula where not retained
Backlog, overdue, team differences, workload imbalance, high-priority pending, recurring categories and equipment patterns	Exact unsupported counts, rankings or percentages
Power BI dashboard built from scratch	Exact historical dashboard layout or page names
Approximately 3-4 hours weekly reporting effort eliminated	Annualized cost savings or named-person time saving

12. Interview-Ready Project Summary

Summary: I worked as a Business Analyst and Data Analyst on a maintenance operations engagement involving approximately 10,000 historical records across 20 teams. The Maintenance Manager was manually consolidating updates and needed better visibility into completion, overdue work and team performance. I helped define the reporting requirements and business rules, cleaned and structured the data, analyzed maintenance performance and built the Power BI dashboard from scratch. The analysis surfaced a 92% overall completion rate along with backlog, overdue work, workload imbalance, high-priority pending work, recurring categories and equipment-related patterns. The reporting solution eliminated approximately 3 to 4 hours of weekly manual reporting effort.